Minimax Bounds Algorithms for MAB

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Multi-Armed Bandit Problem

- A decision maker repeatedly chooses among K actions (arms)
- Each arm gives a reward
- Rewards are unknown
- Goal: maximize cumulative reward over time

Key challenge:

Exploration vs Exploitation

- Explore: try different arms to learn their rewards
- Exploit: play the arm that looks best so far

Why "Multi-Armed Bandit"?

Each arm $=$ one slot machine.

- Each machine has unknown payout
- You want to earn as much money as possible
- You must learn while playing

Adversarial Multi-Armed Bandit

Rewards are not generated from fixed distributions.

- Environment can choose rewards arbitrarily
- Rewards may depend on past actions
- No statistical assumptions

Interpretation:

- Environment may be hostile or strategic
- Past rewards may not predict future rewards

Goal:

Perform well against the worst possible sequence

Stochastic Multi-Armed Bandit

Each arm has a fixed reward distribution.

- Arm i has distribution ν_i
- Rewards are i.i.d. over time
- Mean reward of arm i is μ_i

Interpretation:

- The world is stable
- Past observations help predict future rewards

Goal:

Identify and play the best arm (highest mean)

Stochastic vs Adversarial Bandits

Stochastic	**Adversarial**
Fixed distributions	Arbitrary rewards
i.i.d. rewards	Possibly changing rewards
Learn the best arm	Compete with best fixed arm
Statistical inference	Worst-case guarantees

Adversarial model is strictly more general.

Bandit Interaction Protocol

For $t = 1, \dots, n$:

1. Learner chooses arm $I_t \in \{1, \dots, K\}$
2. Environment generates reward vector

$$g_t = (g_{1,t}, \dots, g_{K,t})$$

3. Learner observes only the reward of chosen arm:

$$g_{I_t,t}$$

This is called **partial feedback**.

Policy

A policy determines how to choose arms.

- Uses past actions and observed rewards
- Outputs probability distribution over arms

$$p_t = (p_{1,t}, \ldots, p_{K,t})$$

Arm I_t is sampled from p_t.

Objective

Total reward of the learner:

$$\sum_{t=1}^{n} g_{I_t,t}$$

But how do we measure performance?
We compare with the best fixed arm in hindsight.

Regret

Regret compares learner to the best arm:

$$R_n = \max_{i=1,\dots,K} \mathbb{E}\left[\sum_{t=1}^{n} g_{i,t} - \sum_{t=1}^{n} g_{I_t,t}\right]$$

Interpretation:

- First term: reward of best fixed arm
- Second term: reward of learner

Goal:

R_n should grow as slowly as possible

Why Regret?

We cannot expect to always play the best arm.
But we want:

$$\frac{R_n}{n} \to 0$$

Meaning:

- Average performance approaches optimal
- Learner eventually behaves optimally

Why EXP3?

- No stochastic assumptions
- Rewards may be adversarial
- Deterministic algorithms fail

Goal:

$$R_n = \max_i \mathbb{E}\left[\sum_{t=1}^{n} g_{i,t} - g_{I_t,t}\right]$$

EXP3: Step 1 — Initialization

Initialize weights:

$$w_{i,1} = 1 \quad \forall i$$

At time t define probabilities:

$$p_{i,t} = \frac{w_{i,t}}{\sum_{j=1}^{K} w_{j,t}}$$

Sample arm $I_t \sim p_t$.

EXP3: Step 2 — Reward Estimation

Observe reward $g_{I_t,t} \in [0,1]$.
Define unbiased estimator:

$$\hat{g}_{i,t} = \begin{cases} \dfrac{g_{i,t}}{p_{i,t}} & i = I_t \\ 0 & \text{otherwise} \end{cases}$$

- Corrects for partial feedback
- High variance

EXP3: Step 3 — Update

Update weights multiplicatively:

$$w_{i,t+1} = w_{i,t} \exp(\eta \hat{g}_{i,t})$$

- Arms with high estimated gain get higher weight
- Randomization ensures robustness

EXP3: Regret Bound

With $\eta \asymp \sqrt{\frac{\log K}{nK}}$:

$$R_n \leq C\sqrt{nK \log K}$$

- Optimal up to $\sqrt{\log K}$
- Gap motivates INF

Minimax Regret: What Is Possible?

Question:

What is the smallest regret any algorithm can guarantee?

We study the **minimax regret**:

$$\inf_{\pi} \sup_{\mu} R_n(\pi, \mu)$$

Main result (stochastic case):

$$\inf_{\pi} \sup_{\mu} R_n(\pi, \mu) \geq c\sqrt{nK}$$

- Holds for *any* policy
- Shows a fundamental difficulty of learning

Lower Bound: High-Level Strategy

Key idea: **confuse the learner**.

- Construct two bandit instances that are:
  - statistically hard to distinguish
  - have different optimal arms
- Any algorithm must fail on at least one

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This forces:

large regret in at least one instance

Lower Bound Construction

Two stochastic bandits:

Instance ν:

$$\mu_1 = \Delta, \quad \mu_2 = \cdots = \mu_K = 0$$

Instance ν':

$$\mu_i = 2\Delta, \quad \mu_1 = \Delta, \quad \text{others } 0$$

- Only *one arm differs*
- Rewards are Gaussian

Why This Works

- Learner must explore all arms
- Some arm is played at most $\frac{n}{K}$ times
- That arm may secretly be optimal

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Tradeoff:

- Explore $\Rightarrow$ regret in ν
- Exploit $\Rightarrow$ regret in ν'

No escape.

Role of KL Divergence

- Learner sees only partial feedback
- Observations under ν and ν' are close

Key inequality:

$$P_\nu(A) + P_{\nu'}(A^c) \geq \frac{1}{2} e^{-D(P_\nu \| P_{\nu'})}$$

KL divergence controls distinguishability.

Lower Bound Takeaway

Optimizing the gap Δ yields:

$$R_n \geq c\sqrt{nK}$$

- No algorithm can beat this rate
- This is the benchmark for all upper bounds

INF: Policy Class

INF uses a function $\psi : \mathbb{R} \to \mathbb{R}_+$ such that:

- ψ is increasing and C^1
- ψ' / ψ is nondecreasing
- $\lim_{u \to -\infty} \psi(u) < 1/K$
- $\lim_{u \to +\infty} \psi(u) \geq 1$

Normalization (Lemma 3)

For any vector $x \in \mathbb{R}^K$, there exists a unique $C(x)$ such that:

$$\sum_{i=1}^{K} \psi(x_i - C(x)) = 1$$

Moreover:

$$\max_i x_i < C(x) \leq \max_i x_i - \psi^{-1}(1/K)$$

Proof uses the Mean Value Theorem.

INF Algorithm

Maintain estimated cumulative gains $\hat{G}_{i,t}$.
Define probabilities:

$$p_{i,t} = \psi(\hat{G}_{i,t} - C_t) \quad \text{with} \quad \sum_i p_{i,t} = 1$$

Sample $I_t \sim p_t$.

INF: Gain Estimation

Define:

$$\hat{g}_{i,t} = \begin{cases} \dfrac{g_{i,t}}{p_{i,t}} & i = I_t \\ 0 & \text{otherwise} \end{cases}$$

Update:

$$\hat{G}_{i,t+1} = \hat{G}_{i,t} + \hat{g}_{i,t}$$

Theorem 4: INF Optimality

Theorem 4 For any real $q > 1$, the Implicitly Normalized Forecaster with

$$\psi(x) = \frac{1}{K}\left(\frac{9\sqrt{qnK}}{-x}\right)^q + \frac{q^{q/(2q-2)}}{\sqrt{nK}}$$

satisfies

$$\sup R_n \leq \frac{37}{1 - 1/q}\sqrt{qnK},$$

where the supremum is taken over all adversarial environments with rewards in $[0, 1]$.

- Achieves optimal $\sqrt{nK}$ rate
- Removes $\sqrt{\log K}$ gap from EXP3
- No explicit exploration

Stochastic Setting

- K arms with unknown means μ_i
- Rewards are i.i.d. and bounded in $[0, 1]$
- Gaps: $\Delta_i = \mu^* - \mu_i$

Regret decomposition:

$$R_n = \sum_{i:\Delta_i>0} \Delta_i \, \mathbb{E}[T_i(n)]$$

UCB1 Algorithm

For each arm i at time t, define

$$\mathrm{UCB}_i(t) = \hat{\mu}_i(t) + \sqrt{\frac{2\log t}{T_i(t)}}$$

At each round:

- Play arm with largest UCB
- Optimism enforces exploration

Theorem (UCB1)

For all $n \geq 1$,

$$R_n \leq 10 \left(\sum_{i:\Delta_i > 0} \frac{1}{\Delta_i} \right) \log n$$

- Instance-dependent logarithmic regret
- Matches lower bounds up to constants

- Concentration: empirical mean stays near μ_i
- A suboptimal arm i is played only when:
  - its estimate is overly optimistic, or
  - the optimal arm is overly pessimistic
- Both events are exponentially unlikely

This yields:

$$\mathbb{E}[T_i(n)] = O\left(\frac{\log n}{\Delta_i^2}\right)$$

Worst-Case Regret of UCB1

Uniform worst case over all instances:

$$R_n = \Theta(\sqrt{nK \log n})$$

- Extra $\sqrt{\log n}$ factor
- Does not match minimax $\sqrt{nK}$

Alternative Regret Bound Formulation

$$R_n \leq \max_{t_i \geq 0, \sum_i t_i = n} \sum_{i:\Delta_i > 0} \min\left(\frac{10}{\Delta_i} \log n,\ t_i \Delta_i\right)$$

- Shows the tradeoff in allocating pulls
- Each suboptimal arm contributes at most $\frac{10}{\Delta_i} \log n$ regret
- But total pulls are limited to n

The Stochastic Case: Motivation

In the deterministic case $g_{i,t} = 1$ if $i = 1$, 0 otherwise:

- INF policies and EXP3 have expected regret $\geq \sqrt{nK}$
- But we want smaller regret in stochastic settings
- Ideally: $\sqrt{nK}$ worst-case, but much better when gaps are large

Let $\Delta_i = \max_j \mu_j - \mu_i$ measure suboptimality.

MOSS Algorithm Definition

MOSS (Minimax Optimal Strategy in the Stochastic case):
For arm i, after s draws:

$$B_{i,s} = \hat{X}_{i,s} + \sqrt{\frac{\max\left(\log\left(\frac{n}{Ks}\right), 0\right)}{s}}$$

with $B_{i,0} = +\infty$.
At time t, draw arm maximizing $B_{i,T_i(t-1)}$.

- $\hat{X}_{i,s}$: empirical mean after s draws
- $T_i(t)$: number of draws of arm i up to time t

MOSS: Key Intuition

- Similar to UCB1 but with $\log(n/(Ks))$ instead of $\log t$
- Bonus decreases as arm gets more pulls
- Arms drawn more than n/K times use just empirical mean
- Automatically balances exploration and exploitation

Theorem 5: MOSS Minimax Bound

Theorem 5 MOSS satisfies:

$$\sup R_n \leq 49\sqrt{nK}$$

where the supremum is taken over all K-tuple of probability distributions on $[0, 1]$.

- Achieves optimal $\sqrt{nK}$ rate
- Matches lower bound up to constant factor
- Constant 49 can be improved to 5.7 with careful tuning

MOSS Proof: Decoupling Trick

Key idea: separate regret into manageable terms.
Define thresholds $z_k = \mu_1 - \frac{\Delta_k}{2}$ (midpoint between best and arm k).
Define $Z = \min_{1 \leq s \leq n} B_{1,s}$ (lowest index of best arm).
Define $\tau_k = \min\{t : B_{k,t} < z_k\}$ (when arm k's index falls below threshold).

MOSS Proof: Crucial Inclusion

Key observation:

$$\{Z \geq z_k\} \subseteq \{T_k(n) \leq \tau_k\}$$

- If best arm's index stays above z_k
- Then once arm k's index drops below z_k, it can't be played again
- So k cannot be pulled more than τ_k times

This decouples the arms!

MOSS Proof: Regret Decomposition

After algebra, we get:

$$R_n \leq 2n\Delta_{k_0} + \sum_{k>k_0} \Delta_k \mathbb{E}[\tau_k] + n \sum_{k>k_0} \mathbb{P}(Z < z_k)(\Delta_k - \Delta_{k-1})$$

Three terms:

1. $2n\Delta_{k_0}$: crude bound for "small-gap" arms
2. $\sum \Delta_k \mathbb{E}\tau_k$: time to rule out each suboptimal arm
3. $n \sum \mathbb{P}(Z < z_k)(\Delta_k - \Delta_{k-1})$: cost when best arm dips

MOSS Proof: Choosing the Split

Pick threshold:

$$\delta_0 = e^{1/16}\sqrt{\frac{K}{n}}$$

Choose k_0 so that:

$$\Delta_{k_0} \leq \delta_0 < \Delta_{k_0+1}$$

- Arms $1, \ldots, k_0$: tiny gaps (hard)
- Arms $k_0 + 1, \ldots, K$: gaps at least δ_0 (easier)

MOSS Proof: Bounding $\mathbb{E}\tau_k$

$$\mathbb{E}[\tau_k] = \sum_{\ell \geq 0} \mathbb{P}(\tau_k > \ell) = \sum_{\ell \geq 0} \mathbb{P}(\forall t \leq \ell, B_{k,t} \geq z_k)$$

For $\ell \geq \ell_0 \asymp \frac{\log(n/(K\Delta_k^2))}{\Delta_k^2}$:

$$\mathbb{P}(B_{k,\ell} \geq z_k) \leq \mathbb{P}(\hat{\mu}_{k,\ell} - \mu_k \geq c\Delta_k) \leq e^{-2\ell c^2 \Delta_k^2}$$

Summing yields:

$$\Delta_k \mathbb{E}\tau_k \lesssim 1 + \frac{\log(n/(K\Delta_k^2))}{\Delta_k}$$

Using $\Delta_k \geq \delta_0$ and maximization:

$$\sum_{k > k_0} \Delta_k \mathbb{E}\tau_k \leq K + 28.3\sqrt{nK}$$

MOSS Proof: Bounding $\mathbb{P}(Z < z_k)$

$Z < z_k = \mu_1 - \Delta_k/2$ means: at some sample size s, empirical mean of arm 1 dipped unusually low.
Translate to partial sums:

$$\sum_{t=1}^{s}(\mu_1 - X_t) \text{ exceeding a boundary}$$

Control using:

- Maximal inequality (Hoeffding for partial sums)
- Peeling argument (split s into geometric ranges)

MOSS Proof: Peeling Computation

After computations:

$$n \sum_{k>k_0} \mathbb{P}(Z < z_k)(\Delta_k - \Delta_{k-1}) \leq n(\delta_0 - \Delta_{k_0}) + 17.8\sqrt{nK}$$

- "Best arm dips" events do happen
- But total contribution is still $O(\sqrt{nK})$

MOSS Proof: Final Combination

Plugging bounds into decomposition:

$$R_n \leq 48.3\sqrt{nK} + K$$

Two regimes:

- If K large compared to n: trivially $R_n \leq n \leq 49\sqrt{nK}$
- Otherwise $K \leq \frac{1}{49}\sqrt{nK}$, so $48.3\sqrt{nK} + K \leq 49\sqrt{nK}$

Thus:

$$R_n \leq 49\sqrt{nK}$$

Comparison of Algorithms

Algorithm	Regret Bound
UCB1	$\min\left(\sqrt{nK\log n}, \sum_{i:\Delta_i>0}\frac{\log n}{\Delta_i}\right)$
MOSS	$\min\left(\sqrt{nK}, \sum_{i:\Delta_i>0}\frac{K\log(2+n\Delta_i^2/K)}{\Delta_i}\right)$
EXP3	$\sqrt{nK\log K}$
INF	$\sqrt{nK}$

- MOSS achieves best of both worlds
- INF optimal in adversarial setting

Summary: Key Takeaways

- Multi-armed bandits model exploration-exploitation tradeoff
- Two main settings: stochastic and adversarial
- Lower bound: $\Omega(\sqrt{nK})$ regret is unavoidable
- UCB1: $\tilde{O}(\log n)$ instance-dependent, $O(\sqrt{nK \log n})$ worst-case
- EXP3: $O(\sqrt{nK \log K})$ in adversarial setting
- INF: Optimal $O(\sqrt{nK})$ in adversarial setting
- MOSS: Optimal $O(\sqrt{nK})$ in stochastic setting with better instance-dependent bounds

Potential Improvements to EXP3

Issue: High Variance Estimator

EXP3 uses unbiased importance sampling estimator:

$$\hat{g}_{i,t} = \frac{g_{i,t}}{p_{i,t}} \mathbb{1}_{\{I_t = i\}}$$

- Variance can be huge when $p_{i,t}$ is small
- Can destabilize learning and slow convergence

Two Possible Improvements:

Improvement 1: Biased Low-Variance Estimators

Replace unbiased estimator with biased but lower variance alternative:

Weighted Importance Sampling:

$$\hat{g}_{i,t} = \frac{w_{i,t} g_{i,t}}{\sum_j w_{j,t} p_{j,t}} \mathbb{1}_{\{I_t = i\}}$$

- Trade off bias for reduced variance
- Can lead to faster empirical convergence
- Similar to difference between ordinary and weighted importance sampling in RL

Open question: Can we bound the regret with biased estimators?

Improvement 2: Baseline Subtraction

Add a baseline b_t to reduce variance without introducing bias:

EXP3 with Baseline:

$$\hat{g}_{i,t} = \frac{g_{i,t} - b_t}{p_{i,t}} \nVdash_{\{I_t = i\}} + b_t$$

- Baseline b_t can be any value known before choosing action
- Common choice: $b_t = \sum_i p_{i,t} \hat{g}_{i,t}$ (weighted average)
- Preserves unbiasedness: $\mathbb{E}[\hat{g}_{i,t}] = g_{i,t}$
- Reduces variance when baseline correlates with rewards

Effect: Smoother updates, better empirical performance

Open Questions

- Can we achieve both optimal stochastic and adversarial guarantees simultaneously?
- What about contextual bandits?
- How to handle very large action spaces?
- Best constants in minimax bounds?

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